

# Functional Features

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Date          | 17 July 2026                                                                         |
| Project Name  | Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction |
| Maximum Marks | 5 Marks                                                                              |

This section evaluates the functional capabilities, primary features implemented, validation systems, and production capabilities of the Smart Lender AI platform.

## Functional Matrix Summary

| Feature Layer             | Details / Implemented Functions                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicant Data Ingestion  | Asynchronous payload submission capturing applicant income data, credit scores, debt structures, and property parameters securely.                                    |
| Predictive Inference Core | Sub-second evaluation engine feeding processed array metrics through loaded tree-model configurations to produce real-time loan approval scores.                      |
| Data Validation Layer     | Two-layer architectural validation handling out-of-bounds metrics at the client interface level, combined with defensive data parsing loops on the server.            |
| User Feedback Dashboard   | Responsive Bootstrap 5 workspace displaying discrete approved/rejected visual badges alongside exact predictive certainty distributions.                              |
| Privacy Preservation      | Strictly transient in-memory operations parsing payloads inside runtime limits without recording intermediate personal transaction details to persistent disk blocks. |

## Functional Quality & Integration Checklist

| S.No | Criteria / Requirement                                           | Status |
|------|------------------------------------------------------------------|--------|
| 1    | System performs operations asynchronously without page refreshes | Yes    |
| 2    | Real-time mathematical inference takes less than one second      | Yes    |

| S.No | Criteria / Requirement                                                     | Status |
|------|----------------------------------------------------------------------------|--------|
| 3    | Incorrect or malicious inputs are intercepted safely before processing     | Yes    |
| 4    | Interface displays output probability along with the final decision status | Yes    |
| 5    | Backend core structures run independently from visual components           | Yes    |
| 6    | Data vectors are normalized before execution to ensure accuracy            | Yes    |

### Additional Notes / Comments

The application successfully fulfills all key requirements for automated predictive credit evaluations. The modular code layout ensures that expanded third-party features, such as live bank APIs or instant messaging alerts, can be added seamlessly in future updates.